# Knowledgebase Assistant – Concept Guide

## 1. Project Overview

The Knowledgebase Assistant is a governed AI workflow developed using Google Opal for Topic 11.

It helps employees receive answers to routine workplace questions using uploaded Standard Operating Procedure (SOP) documents. It also identifies sensitive HR, legal, and compliance queries that require human review.

## 2. Problem Statement

Employees may need help understanding workplace procedures, IT support instructions, leave processes, and expense reimbursement requirements.

A general-purpose AI assistant may produce unsupported answers or handle sensitive complaints inappropriately.

The Knowledgebase Assistant addresses these risks through governance classification, document-grounded responses, and human escalation routing.

## 3. Main Objectives

- Accept workplace questions from employees.
- Classify queries as OPERATIONAL or SENSITIVE.
- Use uploaded SOP documents to answer operational questions.
- Avoid unsupported or fabricated policy information.
- Route sensitive queries to a human escalation handler.
- Display the appropriate workflow response.

## 4. Core Concepts

### SOP-Based Knowledge Retrieval

The workflow uses uploaded static SOP documents as its knowledge sources. Responses must be grounded in the available documents.

### Governance Classification

Each employee query is classified into one of two categories:

- **OPERATIONAL:** Routine workplace questions, such as leave procedures, IT support, and expense reimbursement.
- **SENSITIVE:** Matters such as harassment, discrimination, legal concerns, fraud allegations, and confidential employee disputes.

### Conditional Routing

The Governance Classifier uses the classification result to select the appropriate workflow destination.

### Human Escalation

Sensitive queries are prepared for human review rather than independently investigated or resolved by the assistant.

## 5. Workflow Components

1. Employee Query – collects the employee's question.
2. Classification Result – classifies the query.
3. Governance Classifier – controls workflow routing.
4. Grounded Research Agent – prepares SOP-based operational answers.
5. Human Escalation Handler – prepares notifications for sensitive matters.
6. Display Response – presents the workflow output.

## 6. Example Knowledge Sources

The demonstration workflow uses three fictional sample SOP documents:

- `employee_leave_policy.md`
- `it_support_sop.md`
- `expense_reimbursement_policy.md`

These are demonstration assets and should not be treated as actual company policies.

## 7. Expected Benefits

- More consistent responses to routine employee questions.

- Reduced risk of unsupported policy answers.
- Clear separation between operational and sensitive queries.
- Human oversight for sensitive matters.
- Improved transparency through source identification.

## 8. Scope and Limitations

The workflow demonstrates SOP indexing, governance classification, grounded answer generation, and escalation preparation.

It does not independently resolve legal or compliance complaints, interpret personal employment contracts, or confirm email delivery without a working email integration.

## 9. Technology

- Google Opal
- AI-powered workflow steps
- Uploaded Markdown SOP documents
- Conditional workflow routing